Selection on Observables I: The Assumption

Stat 256, Spring 2026

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Today: the assumption, its two languages (CI $\leftrightarrow$ backdoor), and stratification as the first estimator. Wednesday: matching, weighting, regression.

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All of these are *estimation* choices. They serve a causal claim only if the underlying *identification assumption* is credible.

Setting

- Units: $i = 1, \ldots, N$
- Treatment: $D_i \in \{0, 1\}$, **not** randomly assigned
- Potential outcomes: Y_{0i}, Y_{1i}; observed $Y_i = D_i Y_{1i} + (1 - D_i) Y_{0i}$
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A covariate X is a **confounder** if it is associated with both D and the potential outcomes Y_d.

The Assumption: Conditional Ignorability

Identification Assumption (Conditional Ignorability / Selection on Observables)

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- No unmeasured confounders
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- "Backdoor criterion satisfied by X" (DAG language)

Read it as: *among units with the same X, treatment assignment is as good as random.*

CI and the backdoor criterion: two languages, one object

In Unit 1 we said: if X satisfies the backdoor criterion relative to (D, Y) in a DAG, the effect of D on Y is identified by adjusting for X:

$$P(Y \mid do(D=d)) = \sum_x P(Y \mid D=d, X=x)\, P(X=x).$$

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Claim: these are the *same statement*, written in different alphabets. Let's see why.

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So "block the backdoors" (DAG side) and "D is as-if randomized within strata of X" (PO side) say the same thing. To make this precise we need to be able to *see* potential outcomes on the graph.

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Take your DAG. Delete all arrows *out of* D. Call this $G_{\underline{D}}$.

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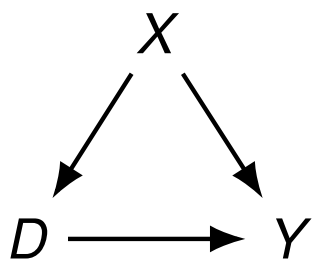

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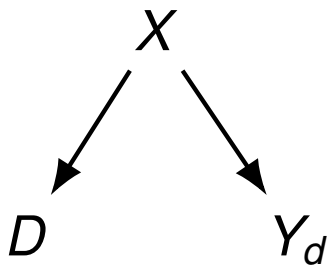

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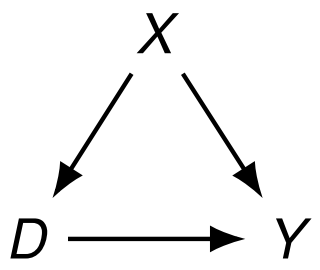

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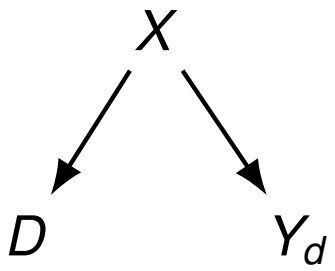

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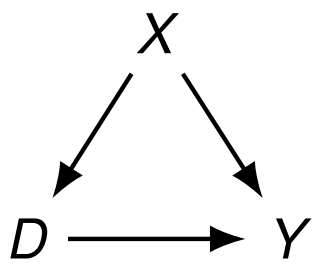

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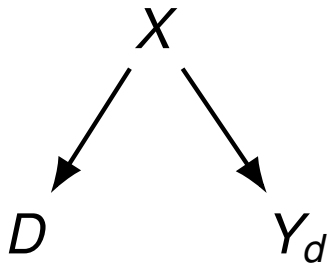

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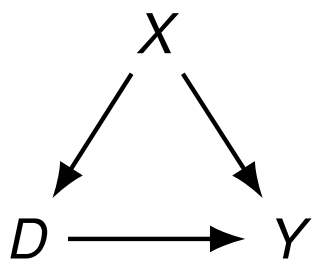

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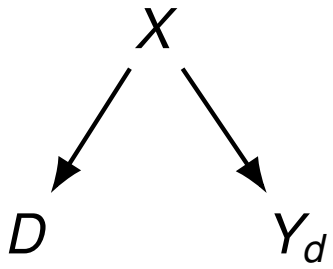

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Thus conditional ignorability is certified.

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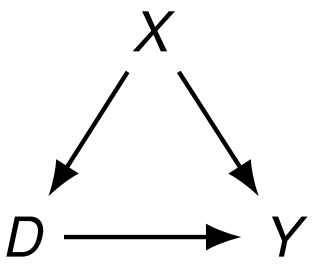

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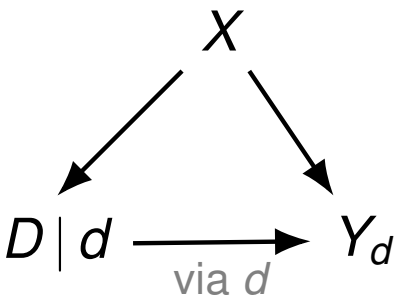

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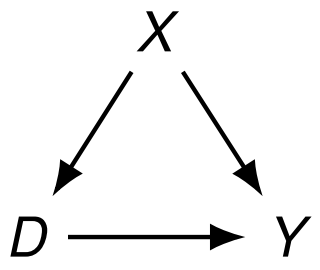

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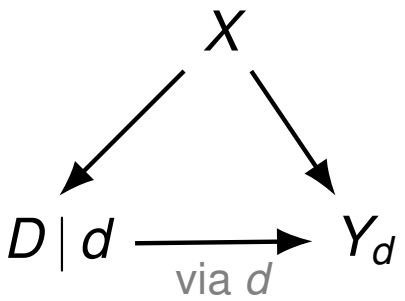

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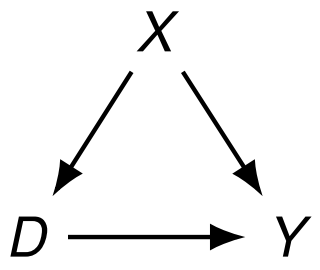

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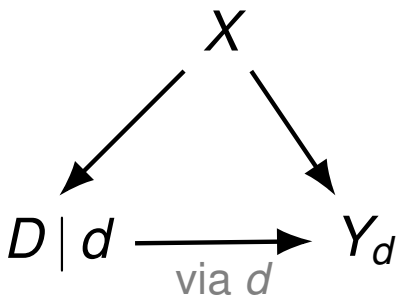

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You can see CI directly as d-separation of D from Y_d in the SWIG:

$$Y_d \perp\!\!\!\perp D \mid X$$

Formal statement, with a nod to the literature

Theorem (backdoor/adjustment criterion $\Rightarrow$ CI). If X satisfies the backdoor criterion relative to (D, Y) in a DAG G, then in any SCM consistent with G,

$$Y_d \perp\!\!\!\perp D \mid X \quad \text{for each } d.$$

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References for more:

- Pedagogical/informal but nice: *What If*, ch. 7.
- Direct, formal: Pearl, Glymour & Jewell (2016), *Primer*, Thm 4.3.1: backdoor-satisfying $Z \Rightarrow Y_x \perp\!\!\!\perp X \mid Z$.
- Also formal: Shpitser, VanderWeele & Robins (2010), Thm 4 states adjustment criterion and ties that to CI on their "counterfactual graph."[1]

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What sorts of stories work?

- **Design**: a conditionally randomized experiment (e.g., block-randomized with different treatment probabilities per stratum). Then CI holds by construction.
- **Observational**: you must *argue* that X captures all common causes of D and Y. Substantive knowledge, not statistical testing.

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CI is **not verifiable from the data**. No amount of checking observed covariates can rule out unobserved confounders.

Common Support / Positivity

CI alone is not enough. Another "identification-like" assumption:

Identification Assumption (Common Support / Overlap)

$$0 < \Pr(D_i = 1 \mid X_i = x) < 1 \quad \textit{for all } x \textit{ in the support of } X.$$

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This is somewhat ill-defined:

- What if X is continuous? Then we are assuming some unclear form of smoothness in $\Pr(D \mid X)$
- "Practical" positivity violations: some strata may lack observed treated or control units, even if they are not zero probability.
- An alternative: abandon common support and worry instead about reasoned uncertainty in each $\mathbb{E}[Y_d|X]$ as we move away from supporting data.

Identification of the ATE under CI

Quantity: $\tau_{\text{ATE}} = \mathbb{E}[Y_{1i} - Y_{0i}]$

[2] Here we are asking for $P(Y \mid do(D = 1)) - P(Y \mid do(D = 0))$ at once, instead of each $P(Y \mid do(D))$.

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The RHS is fully observable (given positivity), so the ATE is **identified**.

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Exactly what you did with the adjustment formula.[2]

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Example: Smoking and Mortality (Cochran 1968)

Unadjusted death rates per 1,000 person-years:

	Canada	U.K.	U.S.
Non-smokers	20.2	11.3	13.5
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Pipe/cigar smokers die at much higher rates. Causal effect of smoking?

Example: Smoking and Mortality (Cochran 1968)

Unadjusted death rates per 1,000 person-years:

	Canada	U.K.	U.S.
Non-smokers	20.2	11.3	13.5
Cigarette only	20.5	14.1	13.5
Cigars/pipes	35.5	20.7	17.4

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Or could something else explain this?

The Confounder: Age

Mean ages by smoking group:

	Canada	U.K.	U.S.
Non-smokers	54.9	49.1	57.0
Cigarette only	50.5	49.8	53.2
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Age is associated with both smoking type and death rate — it is a confounder. Can we adjust for it?

Subclassification Reveals the Effect

Fabricated data inspired by the U.S. figures (death rates per 1,000 person-years):

Age group	Rate (Cig)	Rate (Non)	Effect	N (Cig)	N (Non)
< 45	5	3	$+2$	40	10
45–54	10	5	$+5$	30	20
55–64	20	10	$+10$	20	30
65+	45	23	$+22$	10	40
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The confounding hid an effect of nearly 10 deaths per 1,000 person-years.

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This is the hard part — not the estimator.

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What we understood simply as collider bias before should be visible to you as a violation of CI. Conditioning on some Z with $D \to Z \leftarrow U \to Y$

- changes the distribution of Y_d (because of the U),
- and does so differently for the treated vs. control (because of the D), so you lose the $Y_d \perp\!\!\!\perp D$ you thought you had.

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Next time: matching, propensity scores, weighting, regression — all trying to do the same thing when literal stratification fails.

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- **Don't condition on post-treatment variables.**
- **Once you believe CI**, the estimation question (matching, weighting, regression) is secondary. We turn to that next.

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- See `PresentationAssignment.pdf` on the course site for the topic list.
- Stick around or email me if you want to talk about this or help find group members.